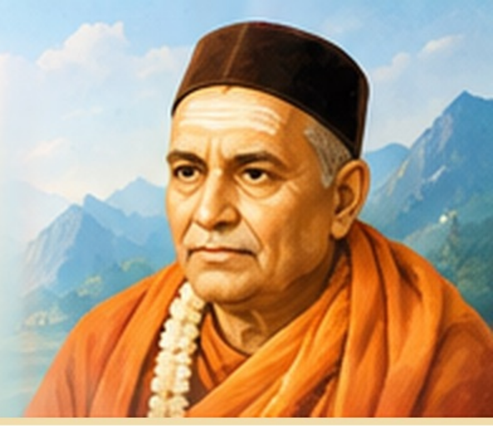
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"Vedic Mathematics is not just a method, it is a way of thinking."



16 Foundational Sutras of Vedic Mathematics

Ancient Wisdom · Modern Speed · Lifelong Skills

1 एकाधिकेन पूर्वेण

Ekādhikena Pūrveṇa

"By one more than the previous one."

Used for: Squaring numbers ending in 5; recurring decimals.

Example: 65^2

$6 \times (6 + 1) = 6 \times 7 = 42$

$5^2 = 25$

$65^2 = 42 \mid 25 = 4225$

Add One. Multiply. Done.

2 निखिलं नवतश्चरमं दशतः

Nikhilaṃ Navataścaramaṃ Daśataḥ

"All from 9 and the last from 10."

Used for: Multiplying numbers close to a base (10, 100, 1000...).

Example: 97 × 96

Deficits from 100: 3 and 4

$97 - 4 = 96 - 3 = 93$

$3 \times 4 = 12$

$97 \times 96 = 93 \mid 12 = 9312$

Subtract. Multiply. Shine.

3 ऊर्ध्वतिर्यग्भ्याम्

Ūrdhva-Tiryagbhyām

"Vertically and crosswise."

Used for: General multiplication of any numbers.

Example: 23 × 14

Vertical: $2 \times 1 = 2$

Crosswise: $2 \times 4 + 3 \times 1 = 11$

Vertical: $3 \times 4 = 12$

2 | 11 | 12 → (carry) 322

Cross. Combine. Complete.

4 परावर्त्य योजयेत्

Parāvartya Yojayet

"Transpose and adjust."

Used for: Division by numbers above a base; solving equations.

Example: $3x + 5 = 2x + 9$

Transpose: $x = (9 - 5) \div (3 - 2)$

$x = 4 \div 1$

$x = 4$ (check: $17 = 17$ ✓)

Reverse. Adjust. Achieve.

5 शून्यं साम्यसमुच्चये

Śūnyaṃ Sāmyasamuccaye

"When the sum is the same, that sum is zero."

Used for: Equations with a common term on both sides.

Example: $(x + 7)(x + 9) = (x + 3)(x + 21)$

Independent terms: $7 \times 9 = 63$

and $3 \times 21 = 63$ → same

So $x = 0$ (check: $63 = 63$ ✓)

Equal. Cancel. Simplify.

6 आनुरूप्ये शून्यमन्यत्

Ānurūpye Śūnyamanyat

"If one is in ratio, the other is zero."

Used for: Simultaneous equations with proportional coefficients.

Example: $12x + 78y = 12$; $16x + 96y = 16$

x-coefficients 12 : 16 = constants 12 : 16

$\Rightarrow y = 0$

$12x = 12 \Rightarrow x = 1, y = 0$

See. Ratio. Solve.

7 सङ्कलनव्यवकलनाभ्याम्

Saṅkalana-vyavakalanābhyām

"By addition and by subtraction."

Used for: Simultaneous equations with interchanged coefficients.

Example: $45x - 23y = 113$; $23x - 45y = 91$

Add: $68x - 68y = 204 \rightarrow x - y = 3$

Subtract: $22x + 22y = 22 \rightarrow x + y = 1$

$x = 2, y = -1$

Add. Subtract. Solve.

8 पूरणापूरणाभ्याम्

Pūraṇāpūraṇābhyām

"By completion or non-completion."

Used for: Quadratic and cubic equations (completing the square).

Example: $x^2 + 6x - 16 = 0$

$(x + 3)^2 - 9 - 16 = 0$

$(x + 3)^2 = 25 \Rightarrow x + 3 = \pm 5$

$x = 2$ or $x = -8$

Complete. Balance. Solve.

9 चलनकलनाभ्याम्

Calana-Kalanābhyām

"By differences and similarities."

Used for: Quadratic equations via the differential (sequential motion).

Example: $x^2 - 5x + 6 = 0$

Differential: $2x - 5$

$2x - 5 = \pm\sqrt{(25 - 24)} = \pm 1$

$x = 3$ or $x = 2$

Differentiate. Relate. Root.

10 यावदूनम्

Yāvadūnam

"Whatever the extent of its deficiency."

Used for: Squaring numbers close to a base (10, 100, 1000...).

Example: 96^2

Deficiency from 100: 4

$96 - 4 = 92$; $4^2 = 16$

$96^2 = 92 \mid 16 = 9216$

Reduce. Square. Join.

11 व्यष्टिसमष्टिः

Vyaṣṭisamaṣṭiḥ

"Part and whole."

Used for: Equations solved by using the average (symmetry).

Example: $(x + 1)^4 + (x + 5)^4 = 82$

Average: let $y = x + 3$

$y^4 + 24y^2 - 25 = 0 \Rightarrow y^2 = 1$

$y = \pm 1 \Rightarrow x = -2$ or $x = -4$ (real)

Part. Whole. Balance.

12 शेषाण्यङ्केन चरमेण

Śeṣāṇyaṅkena Carameṇa

"The remainders by the last digit."

Used for: Recurring decimals of fractions such as 1/7.

Example: 1 ÷ 7

Remainders: 3, 2, 6, 4, 5, 1

× 7 → 21, 14, 42, 28, 35, 7

Last digits: 1/7 = 0.142857...

Remainder. Multiply. Reveal.

13 सोपान्त्यद्वयमन्त्यम्

Sopāntyadvayamantyam

"The ultimate and twice the penultimate."

Used for: Fraction equations whose factors are in A.P.

Example: solve for x

$1/(x+2)(x+3) + 1/(x+2)(x+4)$

$= 1/(x+2)(x+5) + 1/(x+3)(x+4)$

L + 2P = 0: $(x+5) + 2(x+4) = 0$

$3x + 13 = 0 \Rightarrow x = -13/3$

Last. Twice. Zero.

14 एकन्यूनेन पूर्वेण

Ekanyūnena Pūrveṇa

"By one less than the previous one."

Used for: Multiplying by 9, 99, 999...

Example: 777 × 999

$777 - 1 = 776$

$999 - 776 = 223$

$777 \times 999 = 776 \mid 223 = 776223$

Less One. Complete. Win.

15 गुणितसमुच्चयः

Guṇitasamuccayaḥ

"The product of the sum equals the sum of the product."

Used for: Checking multiplication and factorisation.

Example: $(x + 3)(x + 4) = x^2 + 7x + 12$

Put x = 1 (sum of coefficients):

$(1 + 3)(1 + 4) = 4 \times 5 = 20$

$1 + 7 + 12 = 20$ ✓

Check. Verify. Trust.

16 गुणकसमुच्चयः

Guṇakasamuccayaḥ

"The factors of the sum equal the sum of the factors."

Used for: Factorising quadratics and cubics.

Example: $x^2 + 5x + 6$

$= (x + 2)(x + 3)$

Check: $1 + 5 + 6 = 12$

$(1 + 2)(1 + 3) = 3 \times 4 = 12$ ✓

Factor. Check. Confirm.

"These sutras are not merely rules for calculation, but keys to a higher way of thinking."

About Swami Bharati Krishna Tirtha

A visionary scholar, monk and mathematician (1884–1960), Shankaracharya of Govardhana Math, Puri, from 1925 to 1960. He presented these 16 sutras and 13 sub-sutras in his book *Vedic Mathematics* (published 1965), making fast mental calculation accessible to all.

Think Faster
Calculate Smarter
Achieve More